Riemann Sums, Accumulation, and the Fundamental Theorem

Riemann sum estimates from tables and functions, antiderivatives, evaluating definite integrals by the Fundamental Theorem, and accumulation functions

The chain this sheet follows. A Riemann sum estimates accumulated change from finitely many samples. Its limit is the definite integral. The Fundamental Theorem then evaluates that integral exactly from an antiderivative, and read the other way it says the accumulation function $F(x) = \int_a^x f(t)\,dt$ has $F'(x) = f(x)$.

Directions.

- Decide first whether the problem wants an *estimate* (a sum), an *exact value* (an antiderivative), or a *conclusion about a function defined by an integral.* The three alternate.
- Show the sum or the evaluated antiderivative, not just the number. Write $F(b)-F(a)$ explicitly on every FTC evaluation.
- Include units where the problem gives them.

1. Find the general antiderivative:

$$\int \left(4x^3 - 2x\right)\,dx$$

Reverse the power rule term by term and remember the constant of integration.

Answer: ________________

2. A car's velocity $v(t)$, in metres per second, is recorded every 2 seconds:

t (s)	0	2	4	6	8
$v(t)$ (m/s)	5	9	12	14	15

Estimate the distance travelled on $[0, 8]$ using a **left** Riemann sum with 4 subintervals of equal width.

Write out the sum before evaluating it, and state Δt.

Answer: ________________

3. Evaluate exactly:

$$\int_1^3 3x^2\,dx$$

Find an antiderivative first, then write the $F(b) - F(a)$ subtraction before simplifying.

Answer: ________________

4. Define $F(x) = \int_0^x (2t+1)\,dt$.

(a) Find $F(3)$.

(b) In one sentence, say what $F(3)$ measures about the graph of $y = 2t+1$ between $t = 0$ and $t = 3$.

Answer: ____________________

5. Using the same velocity data:

t (s)	0	2	4	6	8
$v(t)$ (m/s)	5	9	12	14	15

Estimate the distance travelled on $[0, 8]$ with a **trapezoidal** sum using all four subintervals. Write the trapezoidal formula with its coefficients before substituting, and say in one sentence why the interior readings are counted twice.

Answer: ____________________

6. Find the general antiderivative:

$$\int (\cos x + 4x)\ dx$$

Handle the trigonometric term and the power term separately.

Answer: ________________

7. Evaluate exactly:

$$\int_0^2 \left(x^3 - 4x\right)\ dx$$

(a) Write $F(2) - F(0)$ and simplify.

(b) Your answer is negative. Explain in one sentence what that says about the graph of $y = x^3 - 4x$ on $[0, 2]$, and why it is not a contradiction for an "area" calculation.

Answer: ________________

8. Let $g(x) = \int_1^x f(t)\,dt,$ where $f(t) = 3t^2 - 6t.$

(a) Find $g(4)$. Put this value on the answer line.

(b) Find $g'(3)$. Do this *without* integrating, and name the theorem that lets you.

(c) Is g increasing or decreasing at $x = 3$? Justify from your answer to (b).

Answer: ____________________

9. Let $f(x) = x^2 + 1$ on the interval $[0, 2]$.

(a) Estimate $\int_0^2 f(x)\,dx$ using a **right** Riemann sum with $n = 4$ subintervals. The four right-endpoint values of f are 1.25, 2, 3.25 and 5. Put the estimate on the answer line.

(b) The exact value of the integral is $\frac{14}{3}$. Is your estimate an overestimate or an underestimate?

(c) Justify your answer to (b) from the behaviour of f on $[0, 2]$, not from the two numbers. Your argument should work for *any* increasing function.

Answer: __________________

10. Challenge. Water flows into a tank at the rate

$$r(t) = 4t + 3 \quad \text{gallons per minute,} \qquad 0 \le t \le 6.$$

At time $t = 0$ the tank already holds 10 gallons. Let $A(t)$ be the number of gallons in the tank at time t.

(a) How many gallons enter the tank during the first 6 minutes? Set up and evaluate a definite integral, and put the value on the answer line.

(b) How many gallons are in the tank at $t = 6$?

(c) Explain, using the Fundamental Theorem, why the integral in part (a) gives the *change* in the amount rather than the amount itself, and why the initial 10 gallons has to be added separately. Write $A(t) = A(0) + \int_0^t r(u)\,du$ and say what each piece contributes.

Answer: ________________